# Abdullah Naveed

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## Education

**University of Florida** Aug 2025 – Expected May 2027
*Ph.D. Candidate in Computer Engineering | GPA: 3.93/4.00* *Gainesville, FL*

**University of Iowa** Aug 2023 – Aug 2025
*Master of Computer Science | GPA: 3.99/4.00* *Iowa City, IA*

**Lahore University of Management Sciences** Aug 2019 – May 2023
*B.S. in Computer Science | GPA: 3.61/4.00* *Lahore, Pakistan*

## Experience

**University of Florida** Aug 2025 – Present
*Graduate Research Assistant, Dependable Systems Lab* *Gainesville, FL*

- Developed **TRACE**, a runtime silent-error detector for LLM inference, achieving **95.3% average detection coverage** across 15 LLMs and 3 datasets with **0.56% runtime overhead**; accepted at SC26.
- Characterize hardware-fault effects in LLM inference and adapted models; evaluate reliability and protection costs. Advised by **Prof. Guanpeng Li**.

**Argonne National Laboratory** Jun 2025 – Present
*Visiting Graduate Researcher* *Remote*

- Develop **scientific data compression and data systems** with Dr. Sheng Di and Prof. Guanpeng Li, addressing compression throughput, data fidelity, and data movement.
- Design and evaluate **CPU, GPU, and FPGA acceleration** through parallel computing and hardware–software co-design.

**University of Iowa** Aug 2023 – Aug 2025
*Graduate Research Assistant* *Iowa City, IA*

- Analyzed hardware-fault effects in scientific and HPC applications using **LLVM, fault injection, and experimental evaluation**; compared compiler and runtime error detection and reliability–performance trade-offs.

**Undergraduate Research Collaboration** Aug 2022 – May 2023
*LUMS · University of Arizona · SRI International*

- Developed a **Python and Docker benchmarking framework** for container debloaters, evaluating attack surface, vulnerabilities, image size, and correctness; **published at IEEE SecDev 2023**.

## Technical Projects

- **Multimodal Fake News Classification** — developed text–image classifiers with **BERT, CNN models, and multimodal fusion**; compared accuracy, precision, recall, and F1.
- **LUMS Mobile App** — built a cross-platform student application using **React Native, JavaScript, and Firebase** for deadline management, GPA tracking, communication, and real-time data.

## Publications

- *Low-Overhead Silent Error Detection for Large Language Model Inference at Scale*. SC26. **Accepted; to appear.**
- *Z-SPDK: Accelerating Lossy Compressed I/O on NVMe-SSDs*. SC26. **Accepted; to appear.**
- *Evaluating Container Debloaters*. IEEE SecDev, 2023.

## Technical Skills

**Programming:** Python, C, C++, JavaScript, SQL, Bash, Git
**Machine Learning / Data:** PyTorch, Hugging Face Transformers, TensorFlow/Keras, scikit-learn, OpenCV, Pandas
**Systems / HPC:** LLVM, Linux, Docker, MPI, OpenMP, Slurm, fault injection, parallel computing
**Hardware:** Chisel, Verilog/RTL, RISC-V, Chipyard, Verilator, Vivado
**Software Development:** React, React Native, Firebase, HTML/CSS